Heuristic Evaluation Synthesis

Across all three evaluations, we had **78 violations.** Violations were spread very evenly across all 10 categories. However, the most common violations were:

- H1: Visibility of System Status (12): Commonly due to the system not telling the user that certain elements were completed (like the learning loop) or reset
- H4: Consistency & Standards (12): Commonly due to using the same Orange Pill design to represent different types of inputs
- H6: Recognition not Recall (11): Commonly due to a lack of placeholders

Severity Breakdown

- Level 4: 11 Violations
- Level 3: 28 Violations
- Level 2: 34 Violations
- Level 1: 5 Violations

This suggests that most of our violations were non-cosmetic usability errors that had a small-to-medium impact on user experience, in addition to a moderate amount of severe usability violations

- Most **severe** violations concerned users being unable to access previous or new information
- Most **major** violations concerned the system not informing the user of a state change
- Many **minor** violations concern design implications, unclear labels, or missing explanations, among other things

Level 4 Violations

- **Tapping anywhere advances the conversation:** We elected to *ignore* this violation, since this behavior was specifically chosen for several reasons:
  - Tapping is how the prototype simulates writing
  - Since there isn't an actual AI responding, we had to make some tradeoffs on how to represent AI-Human interaction
    - Figma also limits what we can do
  - We decided to give the user full control over showing both the AI and User responses for transparency
  - Since the user can potentially write (and tap) anywhere on their side, and they also control the AI, it made sense to progress the conversation by allowing tapping anywhere
- **The Learning Loop doesn't strongly indicate its completion:** we both made the final human response more obvious that it was done, and created a modal that pops up to explicitly signal the end of

the session

    - These changes more closely mirror how a human would actually "end" the session (although these sessions can't end), and the pop-up is the strongest possible indication of completion
- **The system doesn't show users that they can scroll right to the User Space:** to address this, we included the AI/User Space scrolling in the onboarding manual (to explicitly and clearly inform the user) and also provided a hint telling the user to scroll right after the first AI response (to remind the user that they can scroll)
- **The Open button is always clickable, even when the fields are not filled in:** we chose to *ignore* this violation.
    - Our prototype models a physical device (think Microsoft Surface Duo)
    - Thus, it's always possible for the user to open the physical device, even if they haven't filled in all three inputs
        - In this case, the device would have to show an error screen
    - The Open button is designed to mimic this behavior insofar as the Open button is akin to opening the foldable
    - So, we chose to show an error screen since opening is always possible
- **The Favorite Learning Tools section shows blank lines with no hints:** this fix was pretty simple: add placeholders over each blank line that suggest the types of things that users can enter
    - This intuitively and immediately suggests what sort of elements the user can write in
- **There's no visual connection between associated AI questions and User responses:** designing a fix for this was harder because of the way that our product is designed
    - Ultimately, we went with a simple solution: number associated AI questions and User responses
    - While it's not sophisticated, it easily and immediately tells the user what to expect
- **Non-Obvious Text Entry Spot in My Response:** we agreed with this one, so we added a simple placeholder in My Responses when no text has been added yet, informing the user that they may write there – clean and simple!
- **When the AI Space fills up, it suddenly and without warning clears the page:** to fix this, we did two things:
    - Added a warning that the page would reset soon
        - Upfront transparency about what will happen avoids confusion
    - Added navigation buttons between the old and new pages to allow the user to see their entire session history
        - The button also helps inform the user about what happened (that a new, indexable page was added after the first page), avoiding confusion

Level 3 Violations

- **The concept of "AI Space" and "User Space" is jargon-like:** we agree, so we renamed AI Space/User Space to "AI Feedback"/"My Responses" – much more personalized and self-obvious
- **Clicking X immediately exits the session with no confirmation/information:** it's understandable how this can cause confusion and worry, so we added a pop-up modal that confirms if the user wants to actually exit or stay (while informing that the session was saved)
    - This not only informs the user but also gives them a safety layer on exiting – increased control!

- **The user must remember what the AI said while scrolling between the AI Space and the User Space:** that's the whole point! The user needs to understand and *remember* what the AI says (we chose to *ignore* this)
- **There is no in-app help explaining how to use the two-screen swiping model:** we addressed this by explaining it in both the initial main page (through a hint) and through our onboarding guide
- **On the Closed Page, there is no indication of which prompts have been completed:** we *ignored* this one, because it's self-evident that a prompt is completed when it has text in it
    - Just to be safe, we explained this in our onboarding guide to reduce confusion
- **During the learning loop, there is no progress indicator showing how far into the conversation the user is:** we *ignored* this because sessions have no set limit, so there's no way to measure "progress"
- **During the learning loop, there is no way to go back and review a previous AI or User response once it has been scrolled past or cleared:** this is a critical flaw of our design, so we added navigation buttons to go back and forth
- **The Closed Page has no placeholders:** we added placeholders that tell the user where/what to write
    - The user is immediately shown how to fill out the page
- **When mid-scroll between AI Space and User Space, content from both panels is visible simultaneously, creating clutter:** the user has the power to scroll past this mid-scroll view (and why wouldn't they?), so we elected to *ignore* this. There's not much we can do within Figma to address this either
- **The orange pill shape is used for two completely different interactive elements across the prototype:** on the Closed page, we removed the orange pill and replaced it with an underline
    - This both solves the inconsistency of the orange pill's purpose and creates consistency with the rest of the page, reducing confusion across the board
- **Once a Favorite Learning Tool has been added to the Settings page, there is no visible way to remove it:** we didn't even think about this! To address this, we added an X button next to each tool to give the user the option to remove the learning tool
    - The X button is intuitive and obvious to all users
- **After closing Settings with the X button, there is no confirmation of settings saving:** we didn't consider this either, so we added a short modal that informs the user that their settings were saved
- **User-entered text is displayed in red throughout the prototype across all tasks and screens:** this is *wrong* – user text is distinctly orange and not red, which does not carry the same connotations

Conclusion

Ultimately, our evaluators appreciated our project and found the concept to be provoking. However, there were several design trends identified by the evaluators that undermined the product. An example of this was an inconsistency in the use of Orange Pills to represent different types of buttons. Another example was that once the conversation fills the page, there's no way to return to the previous conversation. Several violations were also either a misinterpretation/misunderstanding of the project or outright wrong – we chose to ignore these. But most of the violations we found were helpful and corrected violations that we overlooked during initial development.